

雙和急診

診斷導向止痛策略參考手冊

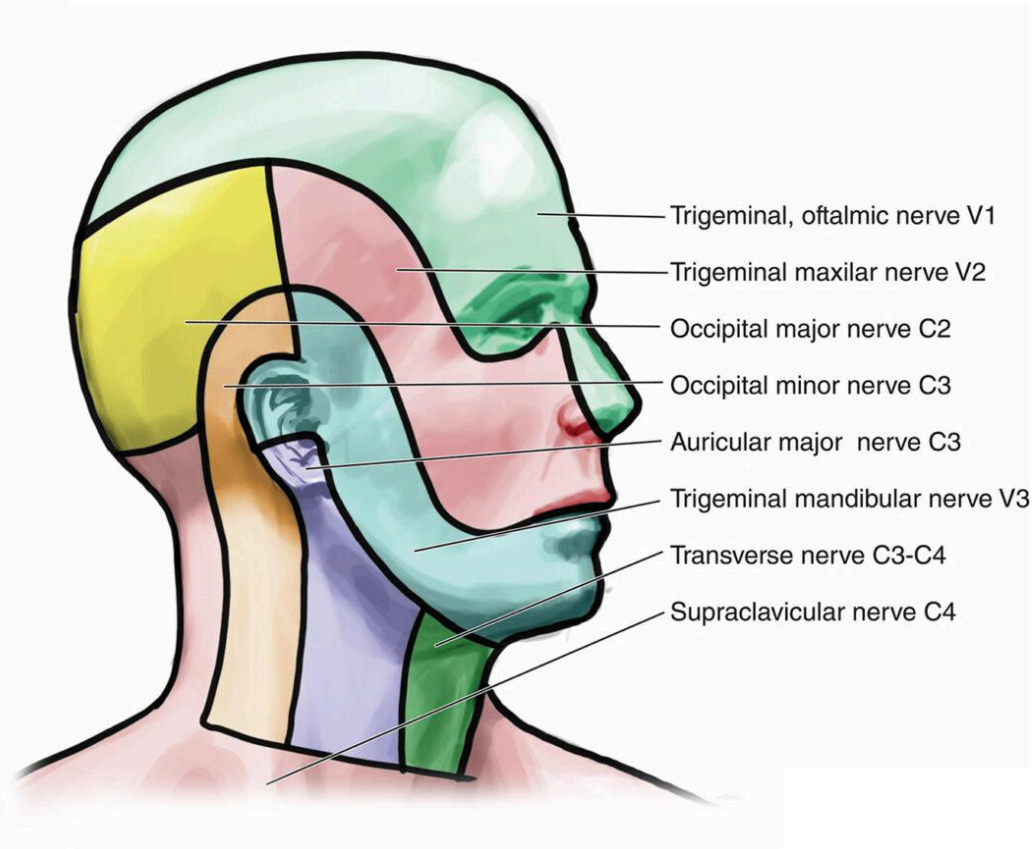
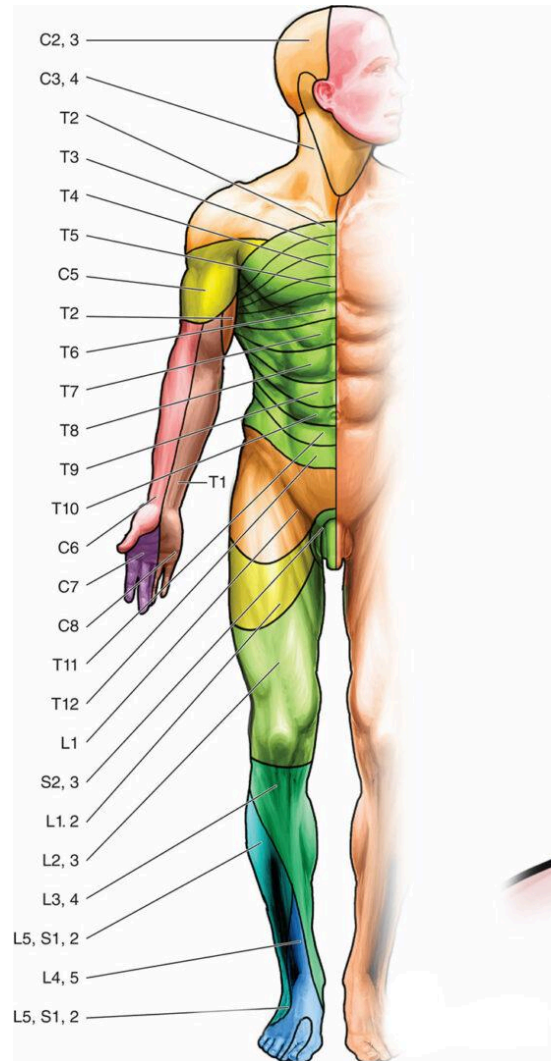
Nerve Block Handbook

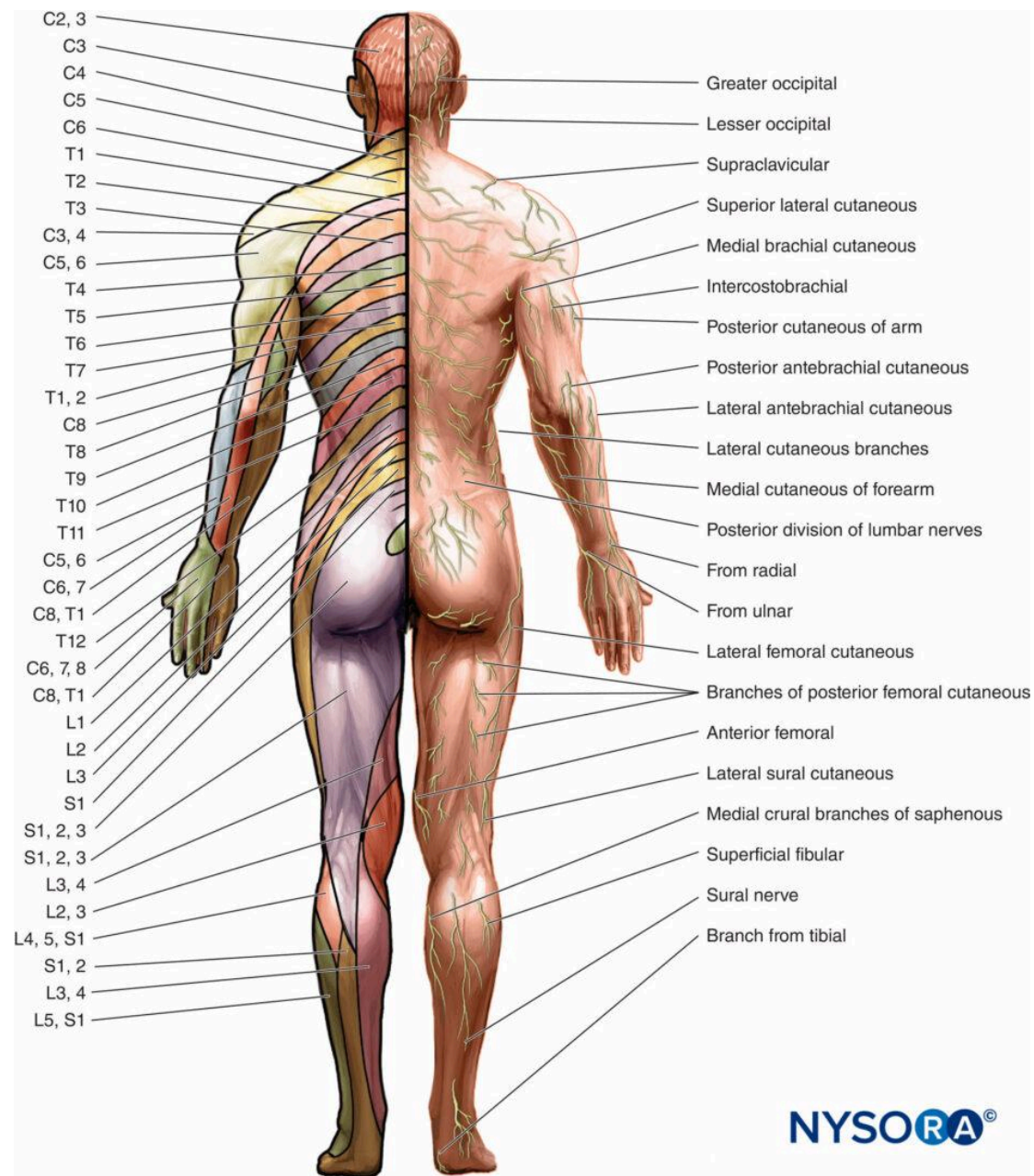


雙和急診醫學科整理

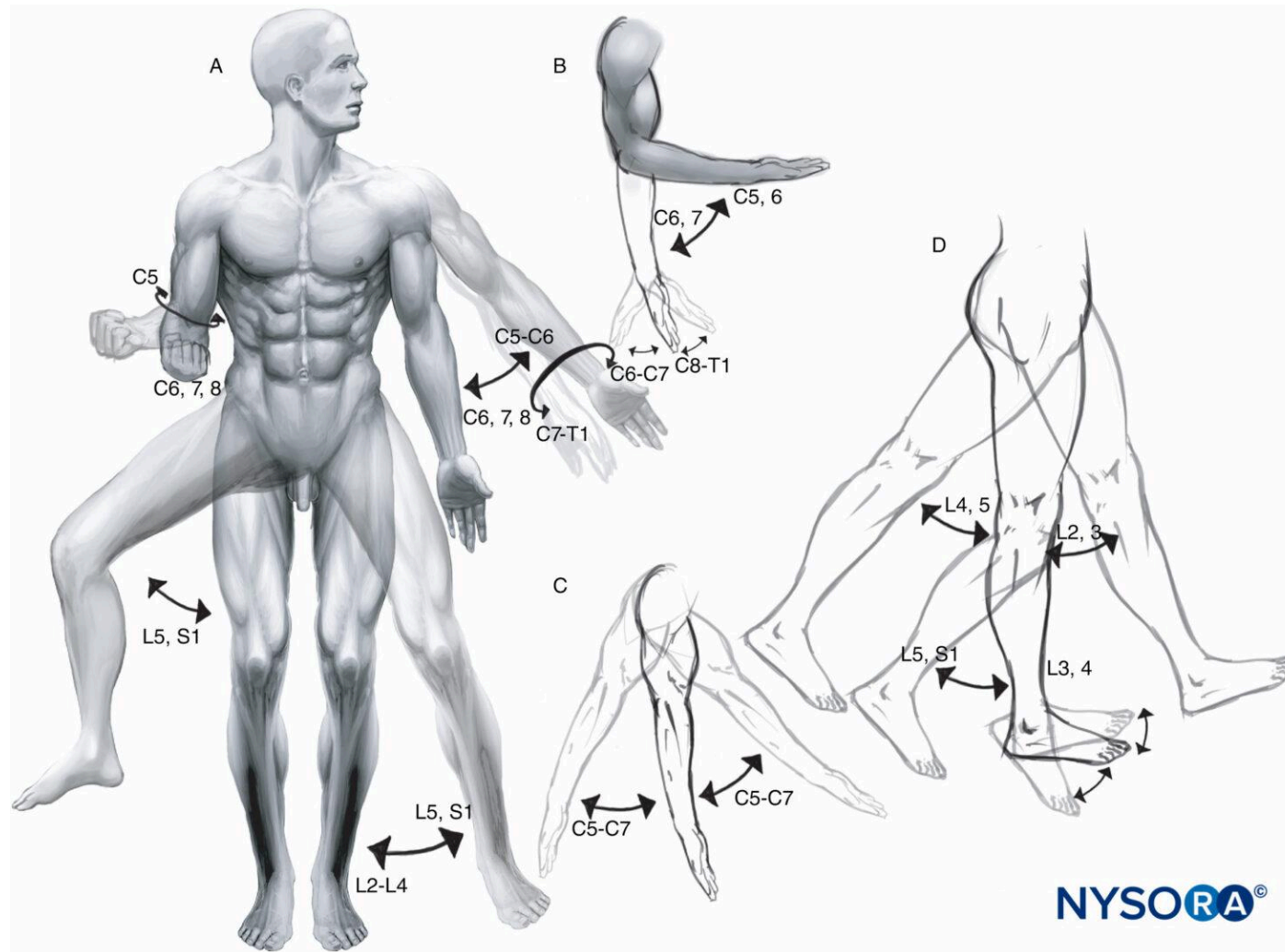
參考資料: POCUS Academy, NYSORA

Dermatome

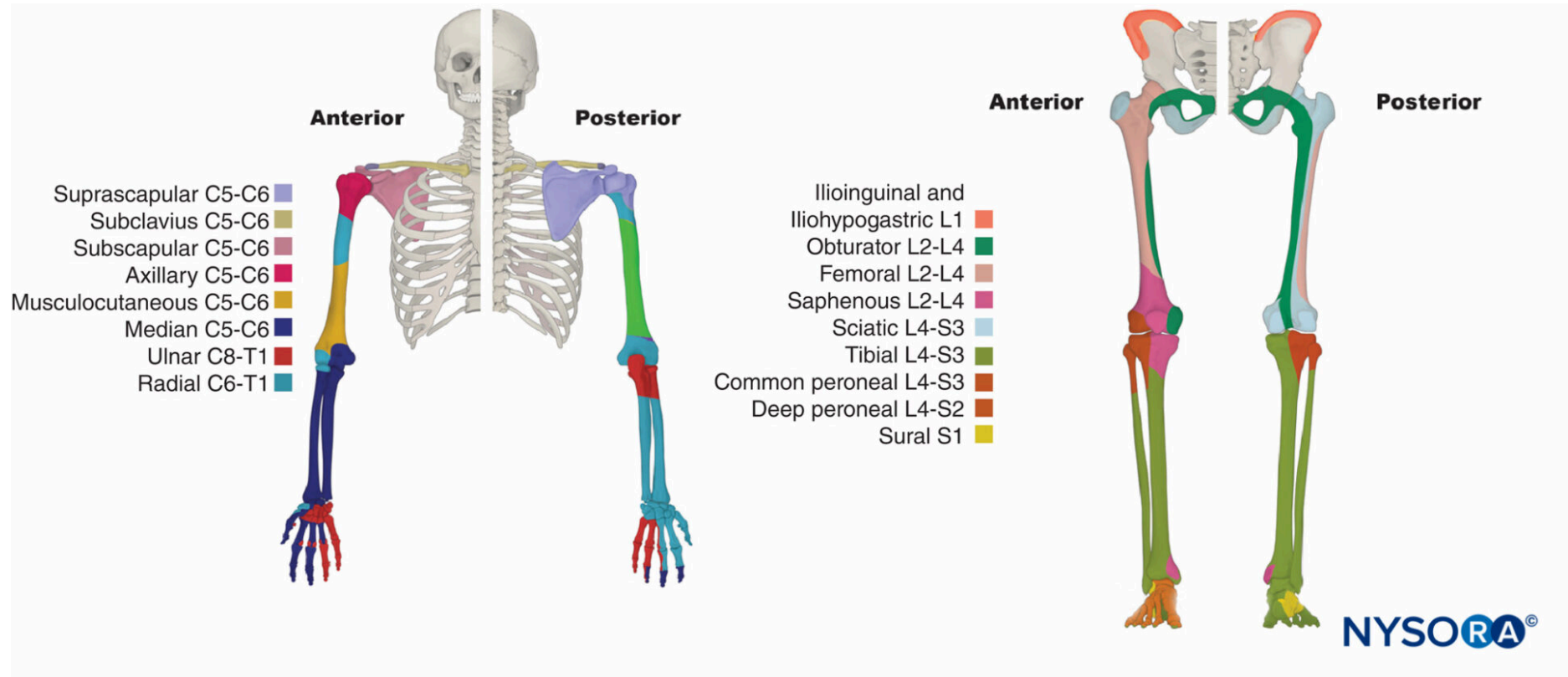




Myotome



Osteotome



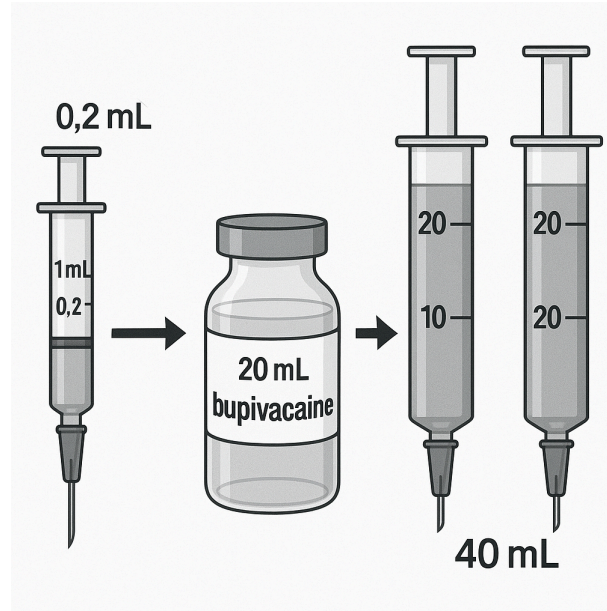
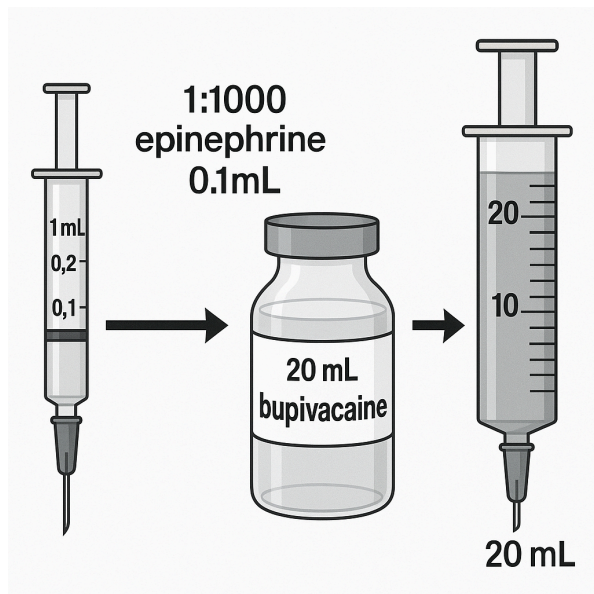
神經阻斷藥物介紹與比較

特性	Lidocaine	Bupivacaine	Ropivacaine
分類	胺類	胺類	胺類
起效時間	快(2–5 分)	慢(5–10 分)	慢(5–10 分)
作用時間	中等(1–2 小時)	長效(4–8 小時)	長效(4–6 小時)
蛋白結合率	約 65%	約 95%	約 94%
脂溶性	中等	高	中高
本院劑型	400mg/20mL/vial	100mg/20mL/vial (0.5%)	200mg/20mL/vial (1%)
最大劑量(無Epi)	4.5 mg/kg(最大300 mg)	2.5 mg/kg(最大175 mg) 60kg = 1.5 amp Marcaine	3 mg/kg(最大200–225 mg)
最大劑量(+Epi)	7 mg/kg(最大500 mg)	3 mg/kg(最大225 mg)	4 mg/kg(最大250–300 mg)
心臟/中樞神經毒性	中	高	中偏低
常用濃度	1–2%	0.25–0.5%	0.2–0.75%
臨床應用	局麻、神經阻滯、硬膜外	硬膜外、神經阻滯、術後止痛	與bupivacaine用途相似，但毒性更低

調和劑 Adjuvants

1. Epinephrine

- a. 用途: 減緩吸收, 延長藥效, 增加最大劑量
- b. 劑量: 1: 200k 至 1:400k
- c. 泡法: 依據最後稀釋完的溶液量, 決定一開始加多少 epinephrine (1:1000) 進 Bupivacaine/Ropivacaine 原瓶 20mL 當中



2. Dexamethasone

- a. 用途: 抗發炎, 常用在 Caudal block, 小型研究也指出可以延長藥效 (perineural 跟 intravenous 效果差不多)
- b. 劑量: 5mg = 1 amp = 1mL

3. Triamcinolone

- a. 用途: 抗發炎, 常用在肌肉拉傷, 筋膜拉傷, 關節腔注射
- b. 劑量: 40mg = 1 amp = 1mL

上肢

外傷診斷	Nerve Block	Volume	Nysora
Finger laceration/dislocation/fracture	Digital block - traditional - palmar approach	3 mL	https://www.nysora.com/techniques/digital-block/
Metacarpal/carpal bone fracture	Forearm median nerve Forearm ulnar nerve Above-elbow radial nerve	5 mL each	https://www.nysora.com/topics/regional-anesthesia-for-specific-surgical-procedures/upper-extremity-regional-anesthesia-for-specific-surgical-procedures/anesthesia-and-analgesia-for-hand-procedures/ultrasound-guided-blocks-elbow/
Distal/shaft radial fracture	Above-elbow radial nerve +/- forearm median nerve	5 mL each	同上
elbow fracture/dislocation 或 radial + ulnar 多根骨頭的骨折	Brachial plexus - Supraclavicular - Costoclavicular	10 mL	https://www.nysora.com/tag/brachial-plexus-block/
Humeral shaft fracture	有很高 radial nerve injury 的可能性 ，不建議打 NB		
Proximal humeral fracture	Brachial plexus - Interscalene - Supraclavicular - Costoclavicular	10 mL	https://www.nysora.com/tag/brachial-plexus-block/ https://www.nysora.com/education-news/tips-for-a-costoclavicular-brachial-plexus-block/
Shoulder dislocation	Brachial plexus - Interscalene - Supraclavicular - Costoclavicular	10 mL	https://www.nysora.com/tag/brachial-plexus-block/ https://www.nysora.com/education-news/tips-for-a-costoclavicular-brachial-plexus-block/
	Axillary nerve + suprascapular nerve block (Motor sparing)	10+10 mL	https://www.nysora.com/education-news/ultrasound-pain-block-tip-of-the-week-suprascapular-nerve-block/ https://www.nysora.com/education-news/shoulder-block-axillary-nerve-block/

胸

外傷診斷	Nerve Block	Volume	Nysora
Clavicle fracture	Clavipectoral fascial plane block	15 mL x 2 (斷端前後)	https://pocusacademy.com/2024/02/29/how-to-perform-an-ultrasound-guided-clavipectoral-block-acep-now/
	Cervical plexus block	5-10 mL (打太多會 hoarsiness)	https://www.nysora.com/techniques/head-and-neck-blocks/cervical/ultrasound-guided-cervical-plexus-block/
	Selective supraclavicular nerve (Include surgical dermatome)	5 mL (打太多會 hoarsiness)	https://pocusacademy.com/2024/10/23/supraclavicular-nerve-selective-block-nysora/
	有時間想要完整Block可以： clavipectoral fascial plane block + cervical plexus block/selective supraclavicular nerve 時間不夠或想要減少入針處： 單打Clavipectoral fascial plane block效果就不錯		
Upper chest wall procedure	PEC I /PEC II block	20 mL	https://www.nysora.com/topics/regional-anesthesia-for-specific-surgical-procedures/thorax/pectoralis-serratus-plane-blocks/#toc_THORACIC-WALL-NERVE-BLOCKS
Anterior ribs fracture	Serratus anterior plane block (SAPB)	20-40 mL	https://www.nysora.com/topics/regional-anesthesia-for-specific-surgical-procedures/thorax/pectoralis-serratus-plane-blocks/#toc_THORACIC-WALL-NERVE-BLOCKS
Posterior ribs fracture	多根肋骨：Erector spinae plane block (ESPB) (止痛範圍大約注射節數上下1.5-2 節，Volume越大節數越多)	25 mL	https://www.nysora.com/erector-spinae-plane-block/

	少根肋骨 : Intercostal nerve block	5 mL each	https://www.nysora.com/topics/regional-anesthesia-for-specific-surgical-procedures/thorax/intercostal-nerve-block/
Sternum fracture	Parasternal PIFB (Pecto-intercostal fascial block) - 安全 - 頭尾的spread較差 - 適合橫向的骨折, 不需要大範圍阻斷	20 mL each	https://pocusacademy.com/2025/03/25/parasternal-nerve-block-for-sternal-fracture/
	Parasternal TTPB (Transversus thoracic plane block) - 靠近肋膜有氣胸風險 - 頭尾的spread好 - 適合胸骨切開術後傷口	20 mL each	https://pocusacademy.com/2025/03/25/parasternal-nerve-block-for-sternal-fracture/

腹

外傷診斷	Nerve Block	Volume	Nysora
Umbilical abscess Umbilical hernia reduction	Rectus sheath block	20 mL each side	https://www.nysora.com/education-news/tips-for-a-rectus-sheath-block-2/
	Anterior/subcostal Transversus abdominis plane (TAP) block	25 mL	https://www.nysora.com/techniques/truncal-and-cutaneous-blocks/truncal-and-cutaneous-blocks/
Inguinal hernia reduction	Ilioinguinal and iliohypogastric nerves block (II&IH block)	10 mL	https://www.nysora.com/techniques/truncal-and-cutaneous-blocks/truncal-and-cutaneous-blocks/
Abdominal wall carbuncle	Transversus abdominis plane (TAP) block	25 mL	https://www.nysora.com/techniques/truncal-and-cutaneous-blocks/truncal-and-cutaneous-blocks/

背

外傷診斷	Nerve Block	Volume	Nysora
Compression fracture	Erector spinae plane block (ESPB) 左右都要打	25 mL each	https://www.nysora.com/erector-spinae-plane-block/
Postherpetic neuralgia			
Lower back pain	Caudal block (濃度建議Bupivacaine < 0.25%, Ropivacaine < 0.2% 避免過麻和下肢無力無法回家的情況)	25 mL	https://www.nysora.com/pain-management/ultrasound-guided-caudal-epidural-injections/
Sciatica			
Lower lumbar (L4/5) fracture			

下肢

外傷診斷	Nerve Block	Volume	Nysora
Calcaneus fracture	Tibial nerve	5 mL	https://www.nysora.com/techniques/ultrasound-guided-ankle-block/
Tarsal/metatarsal fracture	Tibial nerve + deep peroneal nerve (足部骨頭只要這兩條就足夠, 除非有其他skin laceration需要補其餘nerve)	5 mL each	https://www.nysora.com/techniques/ultrasound-guided-ankle-block/
	Popliteal sciatic nerve	10 mL	https://www.nysora.com/techniques/lower-extremity/ultrasound-guided-popliteal-sciatic-block/
Ankle dislocation	Popliteal sciatic nerve	10 mL	https://www.nysora.com/techniques/lower-extremity/ultrasound-guided-popliteal-sciatic-block/
Lateral malleolus fracture	Popliteal sciatic nerve	10 mL	https://www.nysora.com/techniques/lower-extremity/

	(内踝cover不到)		ultrasound-guided-popliteal-sciatic-block/
Medial malleolus fracture	Popliteal sciatic nerve +/- Adductor canal saphenous block	10 mL + 10 mL	https://www.nysora.com/techniques/lower-extremity/ultrasound-guided-popliteal-sciatic-block/ Ultrasound-Guided Saphenous (Adductor Canal) Nerve Block - NYSORA
Tibial/fibular shaft fracture	Popliteal sciatic nerve	10 mL	https://www.nysora.com/techniques/lower-extremity/ultrasound-guided-popliteal-sciatic-block/
Tibial plateau fracture	Femoral nerve block or Adductor canal saphenous block	10 mL	https://www.nysora.com/techniques/lower-extremity/ultrasound-guided-femoral-nerve-block/ Ultrasound-Guided Saphenous (Adductor Canal) Nerve Block - NYSORA
Patellar fracture			
Distal femur fracture			
Femur shaft fracture			
Intertrochanteric fracture			
Femoral neck fracture	PENG	25 mL	https://www.nysora.com/education-news/the-hip-block-new-addition-to-nysoras-web-app/
	Supra-inguinal Fascia iliaca block	40 mL	https://www.nysora.com/topics/regional-anesthesia-for-specific-surgical-procedures/lower-extremity-regional-anesthesia-for-specific-surgical-procedures/ultrasound-guided-fascia-iliaca-block/
Superior pubic ramus fracture	PENG	25 mL	https://www.nysora.com/education-news/the-hip-block-new-addition-to-nysoras-web-app/

LAST處理流程

LAST: SYMPTOMS, DIAGNOSIS & TREATMENT

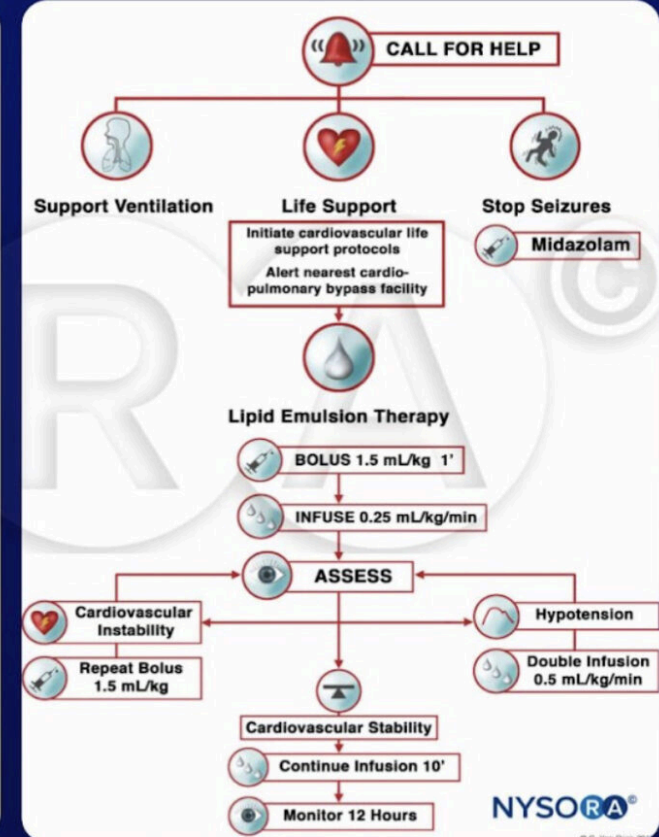


SYMPTOMS OF LAST (LOCAL ANESTHETIC SYSTEMIC TOXICITY)

- CNS more sensitive to LA toxicity than CVS → CNS symptoms precede CVS symptoms
- LAs affect the balance between inhibitory and excitatory pathways in the CNS
- Neurological signs/symptoms range between excitation (early) and depression (late)
- Blockade of the voltage-gated sodium channels of inhibitory cortical neurons in the CNS occurs first → initially LAST = excitatory manifestations
- **Neurological symptoms:** Seizures (70%), agitation (10%), or loss of consciousness (7%)
- **Prodromal (EARLY) symptoms:** Perioral paresthesia, metallic taste, and tinnitus
- **± 40% of LAST** - present as a sudden, rapid-onset seizure, progressing to cardiac arrest
- With high dose of LA or inadvertent IV injection, EARLY (prodromal) CNS symptoms may be absent and the first manifestation could be CVS toxicity (11%)
- **CVS toxicity:** Heart conduction anomalies, ↓ cardiac contractility, ↓ systemic vascular resistance
- **Early-onset ECG changes:** ↑ PR and ↑ QTc, QRS abnormalities (bundle branch blocks) and ↑ ST intervals with/without refractory brady/tachyarrhythmias
- Depression of spontaneous pacemaker activity can rapidly lead to high-degree AV blocks or **asystole**
- Cardiogenic shock and refractory hypotension may follow as a result of ↓ cardiac contractility and vasomotor control disturbances, caused by peripheral vascular ion-channel alterations

DIAGNOSIS

- LAST can occur immediately at the time of injection (*usually accidental intravascular injection*) or up to an hour after it (*due to delayed tissue absorption*)
- Continue monitoring for 30-45 min after injection of large volumes or toxic doses of LAs
- Monitor patients with **any signs of LAST** for 2-6 hrs because cardiovascular depression due to LAs can persist or recur after treatment



Local Anesthetic Systemic Toxicity (LAST)

是指局部麻醉劑(Local Anesthetics, LA)進入系統循環後, 對中樞神經系統和心血管系統產生毒性反應的臨床狀況

危險因子 (Risk Factors)

劑量過大, 注射誤入血管, 肝功能不良(代謝變慢), 年齡極端(老年人嬰兒), 使用高毒性局麻藥(如:bupivacaine), 使用於血流豐富部位(如:頭頸部...)

臨床表現 (Clinical Manifestations)

中樞神經系統 (CNS) 症狀:

早期:口唇麻木、耳鳴、暈眩、視覺模糊、金屬味, 進一步:肌肉顫抖、抽搐、意識喪失, 晚期:呼吸抑制、昏迷

心血管系統 (CVS) 症狀

心律不整(如:心室早期收縮、心室頻脈)、血壓下降、心搏過緩、心搏停止(尤其是bupivacaine)

治療處置 (Management)

停用局部麻醉藥物

急診基本ABC

Lipid Emulsion Therapy(脂質乳劑療法)

1. **1.5 mL/kg** IV推注(最多100 mL)
2. 接續以 **0.25 mL/kg/min** 持續輸注
3. 若無改善, 可重複一次推注並將滴速增至 **0.5 mL/kg/min**

抽搐:可用**benzodiazepines**

心律不整:避免使用**lidocaine**、**procainamide**

心搏停止:按照ACLS標準流程但 Epinephrine 改用原本1/10劑量: 0.1mg/dose

如何預防 (Prevention): 使用最小有效劑量, 分次注射, 反抽檢查(避免誤入血管), 考慮使用超音波導引

神經阻斷病歷紀錄參考

inform the indications, risks and complications of peripheral nerve block
patient/family agreed with nerve block for pain control

Indication: Lt intertrochanteric fracture

Target: Lt femoral nerve

Findings: identified left femoral nerve, femoral artery and vein

Procedure:

1. Antiseptic preparation
2. Total amount of LA: marcaine 0.5% 10ml (with 1:200K epinephrine) / 0.25% 20ml (with 1:200K epinephrine) around target
3. No complication noted during or after the procedure
4. patient tolerated the procedure well
5. Suggestion: wound care
6. NRS: pre-nerve block () > post-nerve block ()

計價碼及計價品項參考

2203021A23	心肌長針23G	健保 5點
2203021A21	心肌長針21G	健保 5點
2210155120	大OP site	自費 23元
F96011C	神經叢阻斷術	健保 1129點
F47051B	末梢神經阻斷術	健保 380點
F96013C	Caudal block	健保 1438點

急診周邊神經阻斷**CHECKLIST**

一、基本資料

病歷號	
姓名	
性別 / 年齡	
日期 / 時間	
主治醫師 / 操作者	

協助者	
疼痛評估 (VAS/NRS)	術前：___ /10 術後：___ /10
Block 類型	☐ 上肢 ☐ 下肢 ☐ 軀幹 ☐ 其他：_____
神經阻斷名稱	
局部麻醉藥選擇	

二、術前準備 Checklist

項目	核對	備註
☐ 病人身份確認(兩項)	☐ 是 ☐ 否	
☐ 知情同意書簽署(急診周邊神經阻斷同意書)	☐ 是 ☐ 否	
☐ 評估過敏史(特別是局麻劑)	☐ 是 ☐ 否	
☐ 評估凝血功能 / 使用抗凝藥物	☐ 是 ☐ 否	
☐ 病人體重以及該局部麻醉藥最大劑量確認	☐ 是 ☐ 否	
☐ 局部麻醉藥物是否需要稀釋	☐ 是 ☐ 否	
☐ 生命徵象穩定	☐ 是 ☐ 否	
☐ 使用鎮靜藥物	☐ 是 ☐ 否	

☐ 器械/超音波機器準備完成	☐ 是 ☐ 否	
☐ 評估執行前perfusion/sensor/motor/NRS	☐ 是 ☐ 否	
☐ 備妥急救設備與藥品	☐ 是 ☐ 否	

三、操作過程紀錄

項目	核對	備註
☐ 確認Block位置與側別	☐ 是 ☐ 否	
☐ 是否有隨時看到針尖	☐ 是 ☐ 否	
☐ 是否有supervisor	☐ 是 ☐ 否	
☐ 是否出現異常感覺 / 疼痛 / LAST症狀	☐ 是 ☐ 否	

四、術後追蹤

項目	核對	備註
☐ 評估執後前perfusion/sensor/motor/NRS	☐ 是 ☐ 否	
☐ 觀察併發症 / LAST	☐ 是 ☐ 否	

五、備註／特殊事項紀錄：
